

Pb:1 The joint probability mass function of two dimensional random variables (X, Y) is

$P(X, Y)$		X		
		0	1	2
Y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

- Q:- 1) Find the marginal distribution functions of X and Y
2) Find the conditional distribution function of X given $Y = 1$.

Soln:-

1) Marginal distribution function of X is $P_X(x) = \sum_y p(x, y)$

" " " " " $P_Y(y) = \sum_x p(x, y)$

$p(x, y)$	x			$P_X(y)$
	0	1	2	
0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$P_X(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Marginal distribution for X

x	0	1	2
$P_X(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

Marginal distribution for Y

y	0	1	2
$P_Y(y)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$

2) we calculate $P(X|Y=1) = \frac{P(x,1)}{P_Y(1)}$

From the marginal distribution of $P_Y(y)$, we have

$$P_Y(1) = \frac{3}{7}$$

We calculate
 $P(x|1)$

if $x=0$, $P(0,1) = \frac{3}{14}$

$x=1$, $P(1,1) = \frac{3}{14}$

$x=2$, $P(2,1) = 0$

Thus the conditional distribution of $P(X|Y=1)$ is

x	0	1	2
$P(x Y=1)$	$\left(\frac{1}{3}\right)\left(\frac{3}{14}\right)$ $\frac{1}{2}$	$\left(\frac{1}{3}\right)\left(\frac{3}{14}\right)$ $\frac{1}{2}$	$\left(\frac{1}{3}\right) \cdot 0$ 0

Pb The joint probability mass function of (X, Y) is

$$P(x, y) = k(2x + 3y), \quad x = 0, 1, 2, \quad y = 1, 2, 3$$

Find all marginal, conditional distributions. Also find the probability distribution of $X + Y$.

soln

$P(x, y)$		x		
		0	1	2
y	1	$3k$	$5k$	$7k$
	2	$6k$	$8k$	$10k$
	3	$9k$	$11k$	$13k$

It is given that $P(x, y)$ is a probability mass function so

$$\sum_x \sum_y P(x, y) = 1$$

$$\Rightarrow 42k = 1$$

$$\Rightarrow k = \frac{1}{42}$$

Marginal distribution

$P(x, y)$		x			$P_Y(y)$
		0	1	2	
y	1	$\frac{3}{42}$	$\frac{5}{42}$	$\frac{7}{42}$	$\frac{15}{42}$
	2	$\frac{6}{42}$	$\frac{8}{42}$	$\frac{10}{42}$	$\frac{24}{42}$
	3	$\frac{9}{42}$	$\frac{11}{42}$	$\frac{13}{42}$	$\frac{33}{42}$
$P_X(x)$		$\frac{18}{42}$	$\frac{24}{42}$	$\frac{30}{42}$	

Marginal distribution of X is

x	0	1	2
$P_X(x)$	$\frac{18}{42}$	$\frac{24}{42}$	$\frac{30}{42}$

Marginal distribution Y is

y	1	2	3
$P_Y(y)$	$\frac{15}{42}$	$\frac{24}{42}$	$\frac{33}{42}$

conditional distribution for X , we calculate $P(X|Y=1)$,

$$P(X|Y=2), P(X|Y=3)$$

$$1) P(X|Y=1) = \frac{P(x_{11})}{P_Y(1)} = \frac{\frac{15}{72}}{\frac{15}{72}}$$

	x	0	1	2
$P(X Y=1)$		$\frac{72}{15} \times \frac{3}{72}$ $\frac{1}{5}$	$\frac{72}{15} \times \frac{5}{72}$ $\frac{1}{3}$	$\frac{72}{15} \times \frac{1}{72}$ $\frac{1}{15}$

$$2) P(X|Y=2) = \frac{P(x_{12})}{P_Y(2)} \Rightarrow \begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(X|Y=2) & \frac{1}{4} & \frac{1}{3} & \frac{5}{12} \end{array} \quad (\text{check!})$$

$$3) P(X|Y=3) = \frac{P(x_{13})}{P_Y(3)} \Rightarrow \begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(X|Y=3) & \frac{3}{11} & \frac{1}{2} & \frac{13}{22} \end{array} \quad (\text{check!})$$

conditional distribution for Y

$$P(Y|X=0) = \frac{P(0,y)}{P_X(0)} \Rightarrow$$

$$P(Y|X=1) = \frac{P(1,y)}{P_X(1)} \Rightarrow$$

$$P(Y|X=2) = \frac{P(2,y)}{P_X(2)} \Rightarrow$$

y	1	2	3
$P(Y X=0)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

y	1	2	3
$P(Y X=1)$	$\frac{5}{24}$	$\frac{1}{3}$	$\frac{1}{24}$

y	1	2	3
$P(Y X=2)$	$\frac{7}{30}$	$\frac{1}{3}$	$\frac{13}{30}$

Probability distribution for $X+Y$

$$P(X+Y=1) = P(X=0, Y=1) = P(0,1) = \frac{3}{72}$$

$$P(X+Y=2) = P(0,2) + P(1,1) = \frac{11}{72}$$

$$P(X+Y=3) = P(0,3) + P(1,2) + P(2,1) = \frac{24}{72}$$

$$P(X+Y=4) = P(1,3) + P(2,2) = \frac{21}{72}$$

$$P(X+Y=5) = P(2,3) = \frac{13}{72}$$

$X+Y$	1	2	3	4	5
$P(X+Y)$	$\frac{3}{72}$	$\frac{11}{72}$	$\frac{24}{72}$	$\frac{21}{72}$	$\frac{13}{72}$